PHYSICS LEC 2

Uniformly Accelerated Motion (Horizontal)

GUIDED EXAMPLES · SOLUTION KEY

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THE FOUR KINEMATIC EQUATIONS

These equations describe motion under constant acceleration. The most efficient strategy is to spot the **missing variable** — the quantity that is neither given nor required — and then choose the equation that omits it.

EQUATION	FORMULA	MISSING VARIABLE
Equation 1	$v = v_0 + at$	Position $(x - x_0)$
Equation 2	$x = x_0 + v_0 t + 1/2\, a t^2$	Final velocity (v)
Equation 3	$v^2 = v_0^2 + 2a(x - x_0)$	Time (t)
Equation 4	$x = x_0 + 1/2 (v_0 + v) t$	Acceleration (a)

GUIDED EXAMPLE 1 Motion of an Air Parcel in a Tube

PROBLEM A parcel of air moving in a straight tube with a constant acceleration of 24.00 m/s² has a velocity of 13.0 m/s at 10:05:00 a.m. (a) What is its velocity at 10:05:01 a.m.? (b) At 10:05:04 a.m.? (c) At 10:04:59 a.m.?

GIVEN

- Acceleration $a = +24.00\ \mathrm{m/s^2}$ (constant)
- Reference time $t = 0$: 10:05:00 a.m.
- Initial velocity $v_0 = 13.0\ \mathrm{m/s}$

REQUIRED

- v at $t = +1.00\ \mathrm{s}$ (10:05:01 a.m.)
- v at $t = +4.00\ \mathrm{s}$ (10:05:04 a.m.)
- v at $t = -1.00\ \mathrm{s}$ (10:04:59 a.m.)

SOLUTION — step-by-step derivation

Step 1. We know v_0, a, and t, and we want v while position x is missing. Choose **Equation 1**.

$$v = v_0 + at$$

Part (a). At $t = +1.00\ \mathrm{s}$:

$$\begin{aligned} v &= 13.0 + (24.00)(1.00) \\ &= 13.0 + 24.0 \\ &= 37.0\ \mathrm{m/s} \end{aligned}$$

ANSWER · Part (a) $v = 37.0\ \mathrm{m/s}$

Part (b). At $t = +4.00\,\text{s}$:

$$\begin{aligned} v &= 13.0 + (24.00)(4.00) \\ &= 13.0 + 96.0 \\ &= 109.0\,\text{m/s} \end{aligned}$$

ANSWER · Part (b) $v = 109.0\,\text{m/s}$

Part (c). At $t = -1.00\,\text{s}$ (one second before the reference time):

$$\begin{aligned} v &= 13.0 + (24.00)(-1.00) \\ &= 13.0 - 24.0 \\ &= -11.0\,\text{m/s} \end{aligned}$$

ANSWER · Part (c) $v = -11.0\,\text{m/s}$

PEDAGOGICAL INSIGHT

The choice of the initial instant $t = 0$ is arbitrary; a negative time simply denotes a state before that chosen reference. The negative result $v = -11.0\,\text{m/s}$ is physically meaningful: at that earlier moment the parcel was moving in the opposite (negative) direction before slowing, reversing, and moving in the positive direction.

GUIDED EXAMPLE 2 Landing Jet on an Aircraft Carrier

PROBLEM A jet lands on an aircraft carrier at a speed of 140 mi/h (≈ 63 m/s). (A) What is its acceleration (assumed constant) if it stops in 2.0 s due to an arresting cable? (B) If the jet touches down at position x_i = 0, what is its final position?

GIVEN

- Initial position $x_0 = 0\,\text{m}$
- Initial velocity $v_0 = 63\,\text{m/s}$
- Final velocity $v = 0\,\text{m/s}$ (fully stopped)
- Time interval $t = 2.0\,\text{s}$

REQUIRED

- Acceleration a in m/s^2
- Final position x_f in metres

SOLUTION — step-by-step derivation

Part (A). We have v_0, v, and t; position is missing, so use **Equation 1** and solve for a:

$$\begin{aligned} a &= \frac{v - v_0}{t} \\ &= \frac{0 - 63}{2.0} \\ &= -31.5\,\text{m/s}^2 \end{aligned}$$

ANSWER · Part (A) $a = -31.5\,\text{m/s}^2$

Part (B). To avoid propagating any rounding in a, omit acceleration and use **Equation 4**:

$$\begin{aligned} x &= x_0 + 1/2\left(v_0 + v\right)t \\ &= 0 + 1/2\left(63 + 0\right)\left(2.0\right) \\ &= 1/2\left(63\right)\left(2.0\right) \\ &= 63\,\mathrm{m} \end{aligned}$$

ANSWER · Part (B) $x_f = 63\,\mathrm{m}$

PEDAGOGICAL INSIGHT

Stress the meaning of the negative sign in $a = -31.5\,\mathrm{m/s^2}$. Ask: does negative acceleration always mean slowing down? Because the initial velocity is positive and the acceleration is negative, the two vectors point in opposite directions, so here the jet does decelerate.

GUIDED EXAMPLE 3 Uniformly Accelerating Object

PROBLEM An object moving with uniform acceleration has a velocity of 12.0 cm/s in the positive x direction when its x coordinate is 3.00 cm. If its x coordinate 2.00 s later is 25.00 cm, what is its acceleration?

GIVEN

- Initial position $x_0 = 3.00\,\mathrm{cm}$
- Initial velocity $v_0 = +12.0\,\mathrm{cm/s}$
- Final position $x = 25.00\,\mathrm{cm}$
- Time interval $t = 2.00\,\mathrm{s}$

REQUIRED

- Acceleration a in $\mathrm{cm/s^2}$

SOLUTION — step-by-step derivation

Step 1. Final velocity v is neither given nor required — it is the missing variable — so use **Equation 2**.

$$x = x_0 + v_0 t + 1/2\,a t^2$$

Step 2. Rearrange to isolate a:

$$\begin{aligned} x - x_0 - v_0 t &= 1/2\,a t^2 \\ a &= \frac{2\left(x - x_0 - v_0 t\right)}{t^2} \end{aligned}$$

Step 3. Substitute the known values:

$$\begin{aligned} a &= \frac{2\left[25.00 - 3.00 - \left(12.0\right)\left(2.00\right)\right]}{\left(2.00\right)^2} \\ &= \frac{2\left[25.00 - 3.00 - 24.00\right)}{4.00} \\ &= \frac{2\left(-2.00\right)}{4.00} = \frac{-4.00}{4.00} \\ &= -1.00\,\mathrm{cm/s^2} \end{aligned}$$

ANSWER $a = -1.00\,\text{cm/s}^2$

PEDAGOGICAL INSIGHT

Reinforce coordinate-system discipline. Writing every algebraic step and every unit exposes why the numerator becomes a negative change $(-2.00\,\text{cm})$, which in turn gives a negative acceleration. Skipping steps is where sign errors creep in.

GUIDED EXAMPLE 4 Electron in a Cathode-Ray Tube

PROBLEM An electron in a cathode-ray tube accelerates uniformly from 2.00 × 10^4 m/s to 6.00 × 10^6 m/s over 1.50 cm. (a) In what time interval does the electron travel this 1.50 cm? (b) What is its acceleration?

GIVEN

- Initial velocity $v_0 = 2.00 \times 10^4\,\text{m/s}$
- Final velocity $v = 6.00 \times 10^6\,\text{m/s}$
- Displacement $x - x_0 = 1.50\,\text{cm} = 0.0150\,\text{m}$

REQUIRED

- Time interval t in seconds
- Acceleration a in m/s^2

SOLUTION — step-by-step derivation

Part (a). Acceleration is the missing variable, so use **Equation 4** and solve for t:

$$\begin{aligned} t &= \frac{2(x - x_0)}{v_0 + v} \\ &= \frac{2(0.0150)}{2.00 \times 10^4 + 6.00 \times 10^6} \\ &= \frac{0.0300}{6.02 \times 10^6} \\ &\approx 4.98 \times 10^{-9}\,\text{s} \end{aligned}$$

ANSWER · Part (a) $t \approx 4.98 \times 10^{-9}\,\text{s} = 4.98\,\text{ns}$

Part (b). Use **Equation 3**, which avoids the time we just computed:

$$\begin{aligned} a &= \frac{v^2 - v_0^2}{2(x - x_0)} \\ &= \frac{(6.00 \times 10^6)^2 - (2.00 \times 10^4)^2}{2(0.0150)} \\ &= \frac{3.60 \times 10^{13} - 4.00 \times 10^8}{0.0300} \\ &\approx 1.20 \times 10^{15}\,\text{m/s}^2 \end{aligned}$$

ANSWER · Part (b) $a \approx 1.20 \times 10^{15}\ \mathrm{m/s^2}$

PEDAGOGICAL INSIGHT

A good place to practise scientific notation and significant figures. Note that $v_0^2 = 4.00 \times 10^8$ is five orders of magnitude smaller than $v^2 = 3.60 \times 10^{13}$, so it barely affects the result — yet keeping it models honest bookkeeping.

GUIDED EXAMPLE 5 The Case of the Impossible Rhinoceros

PROBLEM Why is the following situation impossible? Starting from rest, a charging rhinoceros moves 50.0 m in a straight line in 10.0 s. Her acceleration is constant during the entire motion, and her final speed is 8.00 m/s.

GIVEN

- Initial velocity $v_0 = 0\ \mathrm{m/s}$ (from rest)
- Displacement $\Delta x = 50.0\ \mathrm{m}$
- Time interval $t = 10.0\ \mathrm{s}$
- Reported final speed $v = 8.00\ \mathrm{m/s}$
- Assumption: acceleration a is constant

REQUIRED

- Show the scenario is internally contradictory

SOLUTION — step-by-step derivation

Method A. Compute the displacement implied by the start/end speeds using **Equation 4**:

$$\begin{aligned} \Delta x &= 1/2\left(v_0 + v\right)t \\ &= 1/2\left(0 + 8.00\right)\left(10.0\right) \\ &= 40.0\ \mathrm{m} \end{aligned}$$

But the problem claims 50.0 m — a direct contradiction.

Method B. Compute the final speed required to cover 50.0 m in 10.0 s from rest:

$$\begin{aligned} v &= \frac{2\Delta x}{t} - v_0 \\ &= \frac{2\left(50.0\right)}{10.0} - 0 \\ &= 10.0\ \mathrm{m/s} \end{aligned}$$

This requires 10.0 m/s, not the stated 8.00 m/s.

Method C. Compare the acceleration implied by two different equations:

$$\begin{aligned} \text{Eq. 2}: 50.0 &= 1/2\, a\left(10.0\right)^2 \Rightarrow a = 1.00\ \mathrm{m/s^2} \\ \text{Eq. 1}: 8.00 &= 0 + a\left(10.0\right) \Rightarrow a = 0.800\ \mathrm{m/s^2} \end{aligned}$$

CONCLUSION · Why it is impossible

A single constant acceleration cannot be both 1.00 m/s² and 0.800 m/s². The displacement, final speed, and constant-acceleration conditions over-constrain the motion and contradict one another, so the situation cannot occur.

PEDAGOGICAL INSIGHT

This is a diagnostic problem: each method attacks the same contradiction from a different variable (displacement, final speed, acceleration). Showing all three teaches students that a well-posed kinematics problem must be self-consistent.

GUIDED EXAMPLE 6 Two Competing Toy Cars

PROBLEM At t = 0, one toy car rolls on a straight track with initial position 15.0 cm, initial velocity 23.50 cm/s, and constant acceleration 2.40 cm/s². At the same moment, a second car rolls on an adjacent track with initial position 10.0 cm, initial velocity 15.50 cm/s, and zero acceleration. (a) When, if ever, do the cars have equal speeds? (b) What are those speeds? (c) When, if ever, do they pass each other? (d) Where? (e) Explain the difference between (a) and (c).

GIVEN

- **Car 1 (accelerating)**
- $x_{1,0} = 15.0\,\mathrm{cm}, v_{1,0} = 23.50\,\mathrm{cm/s}$
- $a_1 = 2.40\,\mathrm{cm/s^2}$
- **Car 2 (constant speed)**
- $x_{2,0} = 10.0\,\mathrm{cm}, v_{2,0} = 15.50\,\mathrm{cm/s}$
- $a_2 = 0$

REQUIRED

- (a) time of equal speeds $v_1 = v_2$
- (b) that common speed
- (c) time(s) they pass $x_1 = x_2$
- (d) positions where they pass
- (e) compare (a) and (c) conceptually

SOLUTION — step-by-step derivation

Part (a). Write each velocity function and set them equal:

$$\begin{aligned} v_1(t) &= 23.50 + 2.40\,t \\ v_2(t) &= 15.50 \end{aligned}$$

$$\begin{aligned} 23.50 + 2.40\,t &= 15.50 \\ 2.40\,t &= -8.00 \\ t &= -3.33\,\mathrm{s} \end{aligned}$$

CONCLUSION · Part (a)

Since motion is considered only for t ≥ 0, there is no future instant at which the two cars share the same speed.

Part (b). Mathematically, at $t = -3.33\,\mathrm{s}$ both speeds equal $15.50\,\mathrm{cm/s}$. Physically, for $t \geq 0$, Car 1 is always faster and still accelerating, so their speeds never match after release.

Part (c). Write each position function and set them equal:

$$\begin{aligned} x_1(t) &= 15.0 + 23.50\,t + 1.20\,t^2 \\ x_2(t) &= 10.0 + 15.50\,t \end{aligned}$$

$$\begin{aligned} 15.0 + 23.50\,t + 1.20\,t^2 &= 10.0 + 15.50\,t \\ 1.20\,t^2 + 8.00\,t + 5.00 &= 0 \end{aligned}$$

Apply the quadratic formula:

$$\begin{aligned} t &= \frac{-8.00 \pm \sqrt{(8.00)^2 - 4(1.20)(5.00)}}{2(1.20)} \\ &= \frac{-8.00 \pm \sqrt{40.0}}{2.40} \\ t_1 &\approx -0.698\,\mathrm{s}, t_2 \approx -5.97\,\mathrm{s} \end{aligned}$$

CONCLUSION · Part (c)
Both roots are negative, so there is no physical time (t ≥ 0) at which the cars pass each other.

Part (d). The (past) mathematical intersection points lie on Car 2's line:

$$\begin{aligned} x(t_1) &= 10.0 + 15.50(-0.698) \approx -0.82\,\mathrm{cm} \\ x(t_2) &= 10.0 + 15.50(-5.97) \approx -82.5\,\mathrm{cm} \end{aligned}$$

Part (e). Equal **speeds** means the two position-time graphs have the same slope (same rate of motion), regardless of location. **Passing** means the graphs intersect — the cars occupy the same point in space — regardless of their rates. They are different questions about different features of the graphs.

PEDAGOGICAL INSIGHT
A superb conceptual question. Car 1 starts ahead $(15.0 > 10.0)$ and faster $(23.50 > 15.50)$, while also accelerating, so the slower constant-speed Car 2 can never catch it for $t \geq 0$. The negative solutions describe where the paths would have crossed in the past had the same equations held before release.

GUIDED EXAMPLE 7 The Speeding Car and the Motorcycle Trooper

PROBLEM You are driving at a constant speed of 45.0 m/s when you pass a trooper on a motorcycle hidden behind a billboard. One second after your car passes the billboard, the trooper sets out from the billboard to catch you, accelerating at a constant 3.00 m/s². How long does it take the trooper to overtake your car?

GIVEN
- Car speed $v_{\mathrm{car}} = 45.0\,\mathrm{m/s}$ (constant)
- Trooper acceleration $a = 3.00\,\mathrm{m/s^2}$ (from rest)
- Trooper starts 1.00 s after the car passes the billboard

REQUIRED
- Time t for the trooper to overtake the car

SOLUTION — step-by-step derivation

Method 1. Let $t = 0$ be the moment the trooper starts. The car already had a 1.00 s head start of $45.0\,\mathrm{m/s} \times 1.00\,\mathrm{s} = 45.0\,\mathrm{m}$:

$$\begin{aligned} x_{\mathrm{troop}}(t) &= 1/2(3.00)t^2 = 1.50\,t^2 \\ x_{\mathrm{car}}(t) &= 45.0 + 45.0\,t \end{aligned}$$

Set the positions equal and simplify:

$$\begin{aligned} 1.50\,t^2 &= 45.0 + 45.0\,t \\ 1.50\,t^2 - 45.0\,t - 45.0 &= 0 \\ t^2 - 30.0\,t - 30.0 &= 0 \\ t &= \frac{30.0 \pm \sqrt{(30.0)^2 + 120}}{2} = \frac{30.0 \pm \sqrt{1020}}{2} \\ &\approx \frac{30.0 + 31.937}{2} \approx 31.0\,\text{s} \end{aligned}$$

ANSWER · Method 1 $t \approx 31.0\,\text{s}$ *(measured from when the trooper starts)*

Method 2. Let $T = 0$ be the moment the car passes the billboard; the trooper's elapsed time is $(T - 1.00)$:

$$\begin{aligned} x_{\text{car}}(T) &= 45.0\,T \\ x_{\text{troop}}(T) &= 1/2(3.00)(T - 1.00)^2 \\ 1.50\,T^2 - 3.00\,T + 1.50 &= 45.0\,T \\ 1.50\,T^2 - 48.0\,T + 1.50 &= 0 \\ T^2 - 32.0\,T + 1.00 &= 0 \\ T &= \frac{32.0 \pm \sqrt{(32.0)^2 - 4}}{2} = \frac{32.0 \pm \sqrt{1020}}{2} \\ &\approx 32.0\,\text{s} \end{aligned}$$

ANSWER · Method 2 $T \approx 32.0\,\text{s}$ *(measured from when the car passes the billboard)*

PEDAGOGICAL INSIGHT

Present both methods side by side. Method 1 measures 31.0 s from the trooper's start; Method 2 measures 32.0 s from the billboard — exactly 1.00 s more, the head-start delay. Both place the overtaking at the same point $(x \approx 1439\,\text{m})$, showing the physical event is independent of the chosen time origin.